



Analysing the Operational Performance of Turkish Airports Using Pythagorean Fuzzy Analytic Hierarchy Process and Combined Compromise Solution Methodology

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Received: 01-13-2026

Revised: 06-24-2026

Accepted: 07-07-2026

Citation: H. Fidan, “Analysing the operational performance of Turkish airports using Pythagorean fuzzy analytic hierarchy process and combined compromise solution methodology,” *Int. J. Transp. Dev. Integr.*, vol. 10, no. 3, pp. 617–641, 2026. <https://doi.org/10.56578/ijtdi100303>.



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Abstract: The aviation industry plays a crucial role in the development of international trade, thanks to its positive impact on economic growth, social development, and environmental sustainability. Although there are various metrics for assessing an airport’s economic benefits, evaluating the operational performance of airports is a key and challenging issue due to the complexity of the issues. This study was prepared to measure the operational performance of airports in Turkey and to demonstrate the applicability of the proposed multi-criteria decision-making (MCDM) model. The dataset used in the analysis was obtained as secondary data from the General Directorate of State Airports Authority (DHMI) source. The application analysis considered 5 criteria and 52 alternatives. The Pythagorean fuzzy analytic hierarchy process (PFAHP) was employed to determine the criterion weights, and the combined compromise solution (CoCoSo) method was utilized to rank the airports based on their performance. Sensitivity analysis has verified the consistency and stability of the model used to assess the Turkish airports. Upon examination of the findings, it was determined that the most important criterion was C_3 “Number of Domestic Passengers,” and it was concluded that A_3 “Istanbul” was the highest rated alternative, while A_{52} “Siirt” was the lowest rated. The model we propose demonstrates its applicability for measuring the operational capacity of airports in Turkey. This study makes a methodological contribution to the evaluation processes of operational performance in airports.

Keywords: Airports; Combined compromise solution method; Airport performance assessment; Pythagorean fuzzy analytic hierarchy process; Multi-criteria decision-making model

1 Introduction

Aviation is one of the most prominent industries thanks to the opportunity to travel quickly and demonstrate long-term resilience. According to data from the International Civil Aviation Organization (ICAO) [1], world toll-scheduled passenger traffic increased by an average of 5.79% annually between 2012 and 2019. Global aviation data for 2012 and 2019 [2, 3] and the 2020 Air Transport Action Group (ATAG) report are shown in Table 1. According to global aviation industry data, the aviation industry makes a very important contribution to the world economy.

Table 1. Global aviation data

Process-Year	2012	2019	2020
Passenger traffic (billion people)	3.04	4.5	–
Commercial cargo (million tons)	50.7	56.1	–
Operating income (billion USD)	705.5	840.8	–
Employment (million people)	–	–	87.7
Economic impact (trillion USD)	–	–	3.5
Cargo value (trillion USD)	–	–	6.5

Note: Endashes indicate not recorded/no data.

Air transportation is an important indicator of economic development, particularly in emerging aviation markets [4]. Consequently, airports are linked to economic development due to their contributions at the regional, national, and international levels [5]. In addition to promoting regional growth, airports generate millions of jobs and attract tourists. Many tourists prefer air travel for its advantages, such as speed, comfort, flexibility, and security [6]. Given that air transportation is a vital factor for economic growth, the performance of airports should be thoroughly examined.

In 2019, total air traffic at airports in Turkey was as shown in Table 2 [7].

Table 2. Aviation data in Turkey

Process	Domestic	International	Total
Air traffic (number)	839,894	716,523	1,556,417
Passenger traffic (number)	99,946,572	108,427,124	208,373,696
Cargo traffic (tons)	–	–	4,090,168
Freight traffic (tons)	–	–	1,522,404

Note: Endashes indicate not recorded/no data.

According to the General Directorate of State Airports Authority (DHMI) 2019 report, Turkey ranks 10th in the world for passenger traffic and 8th for cargo traffic. The number of personnel working at 56 airports in Turkey is 205,000, and the aviation industry's turnover is 18.5 billion USD. Among the most important airports are Istanbul, Antalya, Ankara, Izmir, and Adana.

Airports have a central position in air transport. The role of airports can be summarized as follows [8]:

- (1) For travelers, airports are multimodal transport terminals.
- (2) They have become economic centers as they include many commercial activities such as hotels and meeting rooms.
- (3) Airports have a driving role in the economic growth of the region.

Airport operations are numerous and complex, encompassing both landside and airside activities. Local and central governments are seeking various ways to conduct these costly operations effectively and efficiently [9]. Therefore, airports need to evaluate their performance to determine whether they are meeting national or international targets [10]. Although airport operational performance analysis has been studied by many researchers, a consensus has yet to be reached [11]. Additionally, some researchers have developed key performance indicators for the assessment of airports [12].

In parallel with the increasing number of airports and criteria, the significance of these criteria may vary depending on the decision-makers and the multidimensional nature of operational performance. Airports encompass a variety of operations, each of which can serve as a tool for evaluation. Therefore, it is possible to define different criteria for operational assessment. This variability presents a methodological challenge in airport operational performance analysis. Multi-criteria decision-making (MCDM) methods can be a powerful tool for addressing this issue. Through these methods, evaluations are conducted considering conflicting criteria, and the best alternatives are selected based on decision-makers' preferences [13]. Measuring airport operational performance also determines the efficiency levels of airports [14].

Conversely, compared to classical MCDM methods, fuzzy sets help decision-makers address the uncertainty and ambiguity of linguistic terms in evaluation processes. One of the extensions of fuzzy sets is Pythagorean fuzzy sets (PFS), which offer greater flexibility to decision-makers as they represent a generalization of intuitionistic fuzzy sets. Therefore, the Pythagorean fuzzy analytic hierarchy process (PFAHP) is utilized to determine the criteria weights, and the PFAHP-combined compromise solution (CoCoSo) hybrid model is proposed for the first time to rank airports.

(1) The motivation for this study is to develop a novel operational performance analysis tool for Turkey's airports, considering the new performance criteria as a developing country. The rising demand, lack of airport staff, air traffic control delays, and natural disasters exacerbate resource misallocation. There is an urgent need for new approaches to investigate which airports contribute more to the country and to compare their performance with other airports. Therefore, the following research questions are specified in this study:

- (2) RQ1: What are the key criteria utilized for operational performance analysis of airports?
- (3) RQ2: Can the proposed fuzzy MCDM model provide a solution for airport operational performance analysis and ranking?

Due to the irregularity of airport operations and the lack of qualified data during the pandemic, the year 2019 was used as the basis for this study. The DHMI, which is responsible for Air Navigation Services in Turkish airspace, is a public institution that provides all air traffic services within the Turkish civil airspace. The performance criteria evaluated in our study were derived from the DHMI Annual Report 2019. This study aims to assess the performance

of 52 airports actively operating in Turkey according to the defined criteria. The study employed PFAHP and CoCoSo, two innovative MCDM methods. The criteria weights were determined using PFAHP, while the CoCoSo method was applied to evaluate airport performance.

The proposed model provides three significant empirical and methodological contributions to the operational performance analysis of airports. Firstly, the PFAHP–CoCoSo model offers a novel methodological contribution, as it has not been addressed in previous studies on airport operational evaluations. Secondly, by using Pythagorean fuzzy numbers in the pairwise comparison of the criteria, the uncertainties that arise while expressing the preferences of the decision-makers are managed more effectively. Finally, the proposed model presents a clear and concise new framework for the performance analysis of airports. Moreover, the insights gained from the results are considered a managerial contribution of the study for both airport administrators and stakeholders benefiting from airport operations.

Due to the lack of qualitative data regarding the COVID-19 period, comparative analyses have not been conducted on a period-by-period basis; however, general information about this period has been included.

2 Literature Review

This section is divided into three subsections. The first subsection discusses studies related to the performance analysis of airports. The second subsection covers studies that applied the PFAHP and CoCoSo methods. The final subsection addresses research gaps in the performance analysis of airports.

2.1 Literature Review of Airport Performance Analysis

Yu [15] developed a network-based data envelopment analysis (DEA) framework grounded in the slack-based measure approach to assess operational efficiency in serial production systems. The proposed model incorporates quasi-fixed resources such as runway, terminal, and apron capacities, while also allowing for unrestricted intermediate link capacities. Its application focuses on evaluating the performance of airports in Taiwan. Airport operational efficiency is decomposed into production and service efficiency, ensuring that the parameters assessed are evaluated more robustly.

According to Francis et al. [16], benchmarking serves as an important managerial instrument for assessing and enhancing airport performance. Using evidence obtained from interviews and a questionnaire administered to managers of the world's leading passenger airports, the authors examine the prevalence, implementation, and outcomes of benchmarking practices within the airport industry.

Yu [17] examines airport performance in three categories: (1) efficiency and effectiveness, (2) financial performance and service quality, and (3) passenger satisfaction. The DEA method was used in the analyses. Due to the lack of publicly available data, this method has been predominantly applied as the main approach in airport efficiency and effectiveness studies. The author recommends that incorporating data based on physical measurements of airports, such as the number of gates and runways, would yield more reliable results.

Humphreys et al. [18] examined airport performance measurement systems under different ownership models in Europe and the United States. They emphasize the need for airports to focus more on their operational context to evaluate their performance objectively. The study found that many performance metrics used in airport evaluations are often based on quantitative and easily measurable variables. Additionally, it highlights that airport performance measurements are constantly evolving in response to rapid growth in demand and technological innovations.

Humphreys and Francis [19] examine various airport performance measurement methods in Europe and the United States, presenting a comprehensive view of different practices. The study concludes that understanding the operational processes reflected in quantitative performance indicators can enhance the ability of airport planners and managers to facilitate organizational learning and encourage innovation-oriented improvements.

Özdağoğlu et al. [20] analyzed airports based on total passenger, cargo, and flight criteria using the Pivot Pairwise Relative Criteria Importance Assessment–Extended (Piprecia-E), Simple Multi-Attribute Rating Technique (Smart), and Measurement Alternatives and Ranking according to the Compromise Solution (MARCOS) methods to evaluate the world's busiest airports. The authors considered a limited number of criteria, which is less comprehensive compared to those proposed in our study. The analysis conducted using the MCDM approach shows similarities to ours. Ömürbek and Balci [21] evaluated the aviation sector and airports based on the number of arriving passengers, number of departing passengers, amount of cargo carried, flight traffic, commercial aircraft fleet, number of IATA member airlines, number of international airports, and fatalities, utilizing the Entropy and Complex Proportional Assessment (COPRAS) methods for the evaluation of air transport in European Union countries and Turkey. Instead of obtaining expert opinions for the evaluation of these criteria, the authors preferred a more objective evaluation approach. In our study, however, expert opinions were included. Ertunç and Çay [22] conducted a site selection analysis for airport location using the AHP method, based on factors such as population density, meteorological data (temperature and precipitation), altitude, slope information, access to land transportation, and distance to provincial

centers. They employed a single-stage MCDM approach to determine the airport location, whereas our proposed model utilizes a layered integrated MCDM approach instead of a single-stage MCDM.

Bakır and Akan [23] analyzed airports based on queuing time, terminal cleaning, terminal seating areas, terminal signage and orientation, catering, airport shopping stores, Wi-Fi connectivity, and airport staff criteria using the Entropy and Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) methods for the evaluation of service quality at airports. This study performs a performance analysis based on different criteria, as discussed in our research. It differs from our proposed model due to variations in the criteria and alternatives considered, as well as differences in the analysis methods employed. Ömürbek and Akçakaya [24] evaluated the aviation sector and airports based on sales, assets, market value, and number of employees using the Entropy, Multi-Attribute Utility Theory (MAUT), Copras, and Simple Additive Weighting (SAW) methods to analyze Forbes 2000-listed aviation companies. In this study, the authors evaluated companies operating in the aviation sector, analyzing data related to the criteria determined for evaluation using integrated MCDM approaches. The criteria and alternatives considered in this study differ from those in our research, and the opinions of decision-makers were not consulted for the criteria used in their analysis. Ekin and Dinçer [25] demonstrated the performance of airports based on warehouse spare number, mandatory code, turnaround duration, test period, confidence level of parts, aircraft type of parts, number of components on aircraft, active inventory ratio, cost of holding, component ordering cost, and component usage frequency, using the Vİsekriterijumsko KOMPromisno Rangiranje (VIKOR) and Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE) methods to address spare parts inventory problems in the aviation industry. This study differs from ours in terms of the criteria data sets and alternatives considered, as the authors aimed to find solutions to the technical problems of airports.

Özdağoğlu et al. [26] analyzed the aviation sector based on total passenger number, total baggage number, lost baggage rate, and damaged baggage rate criteria using the Best–Worst Method (BWM), multi-attributive ideal-real comparative analysis (MAIRCA), and multi-attributive border approximation area comparison (MABAC) methods to evaluate the performance of airline companies operating from Isparta Suleyman Demirel Airport. In this study, the alternatives considered are airline companies rather than airports. The operational activities of the businesses at the airport in question were taken into account, remaining specific to that airport. It differs from our study in terms of the weighting and ranking methods used. Akdeniz [27] evaluated in-flight service quality based on criteria such as food and beverage, in-flight entertainment, seat comfort, and personnel service using the AHP method to select the best airline operator. Peker and Birdoğan [28] demonstrated airport performance based on criteria including key operational indicators, including parking capacity, runway infrastructure, airport size, workforce level, passenger demand, and cargo value, applying the DEA method for efficiency measurement at Turkish airports. This study considered one method, as the MCDM approach, and the number of criteria was kept limited.

Bolat et al. [29] evaluated airports based on the number of check-in counters, baggage conveyors, passenger boarding gates, runways, terminal size, number of employees, parking lot capacity, total number of passengers, total cargo volume, and commercial flight traffic criteria using DEA and artificial neural network methods for activity forecasting of airports in Turkey. This study is similar to ours; however, it differs in terms of the considered criteria, alternatives, and the MCDM method employed. The criteria related to the total number of passengers, total cargo volume, and commercial flight traffic are consistent with those in our study. Ömürbek et al. [30] evaluated the aviation sector and airports based on flight activity, commercial air traffic, cargo and passenger throughput, sales income, service area, passenger and parking capacities, vehicle fleet size, apron and aircraft capacities, together with the availability of computing and rescue equipment, and personnel, using the DEA method for performance measurement in the airport service sector. Some of the criteria in this study are similar to those in ours (flight traffic, commercial flight traffic, cargo traffic, and passenger traffic), but the analysis method applied differs. Şahin [31] evaluated airports based on workforce size, operating expenditures, terminal capacity measured by area, runway and apron infrastructure, passenger throughput, aircraft movements, and freight-related traffic consisting of cargo, mail, and baggage volumes, utilizing both DEA and the Malmquist total factor efficiency index methods to analyze the financial activities of airports in Turkey. This study employed the MCDM approach and considered the financial-cost aspect of airports for operational evaluation.

According to the literature review, various studies on airport operations and activities have addressed these issues using MCDM approaches. The criteria used in these analyses are generally similar to those considered in our study. However, the weighting and ranking methods, as well as the alternatives evaluated, differ across studies. The issues examined in the literature are directly related to airport operational activities. While some studies incorporate expert opinions, others rely on objective data. Comparative analyses involving multiple MCDM methods are rarely included. In general, airports are evaluated and ranked according to their operational performance based on specified criteria. The reviewed studies did not employ fuzzy MCDM approaches. Although there are similarities among the criteria used, each study also considers different criteria depending on the research scope and expert evaluations. In addition, most studies involve a limited number of alternatives and primarily focus on airports in Turkey. As demonstrated in the case studies, airports can be evaluated from different functional perspectives, and even operational criteria may

vary, as in our study. Therefore, there is no universally accepted set of criteria for airport evaluation; rather, the selection of criteria depends on the researcher's assessment and expert opinions.

A review of previous studies indicates that no prior research has applied the specific MCDM approach proposed in this study. Likewise, both the diversity and the number of evaluation criteria used in our research differ from those employed in earlier studies.

Although numerous studies on performance measurement exist in the literature, research specifically focusing on airport operational performance measurement remains limited. Non-financial indicators are generally considered more appropriate for evaluating performance dimensions that are important to stakeholders, including customers, local governments, regulators, and the wider community. Furthermore, performance measurement practices in airport environments differ from those applied in other related business contexts.

2.2 Literature Review of The Proposed Methods

This section focuses not on airport applications, but rather on examples of the proposed model applied in different fields. The aim is to demonstrate that the proposed model can be effectively used to solve a wide range of decision-making problems.

Ilbahar et al. [32] conducted an occupational health and safety risk assessment using the Pythagorean fuzzy proportional risk assessment, Fine–Kinney, and PFAHP methods. The authors evaluated the alternatives based on a range of factors associated with workplace safety and operational performance, including environmental conditions, personnel management practices, unsafe actions, heavy machinery operations, and construction site supervision. This study demonstrates that the proposed MCDM framework can be successfully applied to problems in different domains and highlights the contribution of the PFAHP approach to complex decision-making processes.

Gul [33] employed the PFAHP and fuzzy VIKOR methods for occupational health and safety risk assessment. The study evaluated risks based on criteria including collision, compression, crushing, electrical hazards, and falling or flying objects, as well as workplace conditions such as wet floors, manual handling, inadequate ventilation, repetitive movements, and machine-related hazards.

Yazdani et al. [34] applied the Decision-Making Trial and Evaluation Laboratory (DEMATEL), BWM, and CoCoSo-G methods to the supplier selection problem. The evaluation process incorporated multiple criteria related to supplier performance and sustainability, including design expertise, greenhouse gas emission levels, delivery performance and flexibility, communication responsiveness, financial stability, pricing policies, and the implementation of environmental management practices.

Karasan et al. [35] used the PFAHP method for landfill site selection by considering environmental, social, economic, and operational criteria. Similarly, Çalık [36] applied the PFAHP and PFTOPSIS methods to the green supplier selection problem using criteria related to delivery, pollution control, production, quality, and environmental performance.

Torkayesh et al. [37] evaluated healthcare sectors using the BWM, Level Based Weight Assessment (LBWA), and CoCoSo methods based on criteria such as the number of doctors, nurses, hospital beds, computerized tomography scanners, magnetic resonance imaging devices, radiotherapy equipment, and mammography machines.

Peng et al. [38] applied the CRITIC and interval-valued CoCoSo methods for intelligent healthcare management evaluation. The analysis considered criteria including body mass index, waist circumference, heart rate, blood pressure, electrocardiogram results, pulmonary X-rays, white blood cell count, haemoglobin, transaminase, and total bilirubin levels.

Deveci et al. [39] employed fuzzy logarithmic and PH'CoCoSo methods to prioritize real-time traffic management approaches based on a set of criteria encompassing economic considerations, public and political factors, environmental impacts, and traffic safety performance.

Ayyildiz and Taskin Gumus [40] extended the Supply Chain Operations Reference (SKOR) model with new metrics using the BWM and PFAHP methods. Their evaluation was based on criteria such as reliability, flexibility, responsiveness, cost, assets, digital technology, and information systems.

Yildiz et al. [41] analyzed the ATM site selection problem using the PFAHP and PFTOPSIS methods under customer, financial, internal, and external criteria.

Ozdemir and Gul [42] investigated the development levels of Nomenclature of Territorial Units for Statistics Level 2 (NUTS-2) regions in Turkey using the PFAHP and Interactive and Multi-Criteria Decision Making (TODIM) methods. The study employed various socio-economic and well-being criteria, encompassing housing, employment-life balance, economic prosperity, healthcare, education, environmental conditions, security, civic engagement, infrastructure accessibility, social interactions, and subjective well-being.

Adar et al. [43] applied the AHP and CoCoSo methods to prioritize industrial wastewater management processes based on technical, cost, and environmental criteria.

Peng and Smarandache [44] evaluated rare earth industry security using the Criteria Importance Through Intercriteria Correlation (CRITIC) and neutrosophic CoCoSo methods considering industry structure security,

organizational security, layout security, policy security, and ecological security criteria.

Unlike the classical AHP method, the fuzzy analytic hierarchy process (FAHP) incorporates fuzzy logic to model uncertainties, subjective judgments, and imprecise information encountered in real-world decision-making problems. Therefore, it provides a more flexible and realistic framework for complex evaluations.

PFAHP can be considered an advanced extension of the classical AHP method, integrating fuzzy logic to better represent uncertainty, ambiguity, and complex human judgments. The main difference between Intuitionistic Fuzzy AHP and Fuzzy AHP lies in the representation of the hesitation degree experienced by decision-makers under uncertain conditions.

The CoCoSo method reduces inconsistencies in decision matrices and produces highly reliable results compared with traditional MCDM methods such as TOPSIS and EDAS. Previous studies have also shown that CoCoSo generates results consistent with methods such as MAIRCA and MABAC.

The integration of the AHP and CoCoSo methods forms an effective hybrid model that is widely used in contemporary decision-making processes, as the strengths of one method help compensate for the limitations of the other. Combining the subjective evaluation capability of AHP, based on expert judgments, with the objective and compromise-based computational structure of CoCoSo enables a more comprehensive and reliable decision-making framework.

AHP is particularly effective for determining criteria weights; however, its application may become more complex as the number of alternatives increases. In contrast, CoCoSo provides a robust and compromise-oriented approach for evaluating and ranking alternatives.

For these reasons, the PFAHP–CoCoSo approach was employed in this study. The study aims to demonstrate that integrating the PFAHP and CoCoSo methods can produce more comprehensive and reliable evaluation results.

The FAHP approach has been extensively applied in MCDM problems due to its ability to model uncertain, imprecise, and linguistically expressed human evaluations within a structured decision framework. However, owing to its complex mathematical structure, the method also has certain limitations. The matrix operations involved in FAHP can become computationally intensive, particularly in large-scale problems, making manual calculations difficult and time-consuming.

Furthermore, during the process of transforming decision-makers' linguistic evaluations (e.g., “moderately important”) into fuzzy numerical values and subsequently converting them into crisp scores, some semantic information may be lost.

Similarly, in certain formulations of the CoCoSo method, normalization procedures may produce value scales that deviate from expected ranges under extreme problem conditions. In highly complex real-world applications involving numerous variables or unusual scenarios, obtaining stable and sufficiently discriminative final rankings may therefore become challenging.

2.3 Research Gaps

Since airports play a significant role in the economic development of countries, numerous studies have investigated airport performance from different perspectives. However, evaluating and comparing airport performance requires flexible, robust, and reliable decision-making tools capable of effectively handling uncertainty and ambiguity.

In this context, this study proposes a novel MCDM framework integrating the PFAHP and CoCoSo methods. To the best of our knowledge, this is one of the first studies to combine PFAHP and CoCoSo approaches for evaluating airport operational performance within a Pythagorean fuzzy environment.

Airports require flexible and robust evaluation models to better assess performance under uncertain conditions. Therefore, PFSs are employed in this study to better represent ambiguity and vagueness in decision-makers' judgments. In addition, the CoCoSo method is used to rank airports according to their operational performance.

The consequences of the COVID-19 pandemic on organizational performance have been extensively discussed in the operational performance literature. However, studies focusing specifically on airport performance remain limited. Moreover, the existing studies exhibit several methodological and structural shortcomings. First, many studies employ limited criteria and alternatives for evaluating airport operational performance. In some cases, data have been collected through surveys conducted with passengers, pilots, or aviation industry managers, which may reduce the reliability and applicability of the findings for performance monitoring and airport selection.

Furthermore, comprehensive methodologies for evaluating airport performance are still limited in both scope and quality. There is also no widely accepted set of criteria that can effectively address real-world decision-making problems related to airport performance assessment. Consequently, the literature lacks a standardized framework for evaluating airport operational performance.

To address these gaps, this study proposes a hybrid PFAHP–CoCoSo MCDM approach for identifying and evaluating the criteria affecting airport operational performance. The proposed framework aims to provide a practical, reliable, and efficient decision-making tool capable of handling uncertainty and ambiguity effectively. The Pythagorean fuzzy approach is employed to determine the evaluation criteria based on expert judgments.

This study primarily focuses on operational processes; therefore, financial and managerial dimensions are excluded from the analysis. In addition, the study offers a practical decision-support framework for policymakers and practitioners responsible for the planning and management of Turkey's transportation and aviation systems. The proposed framework may assist decision-makers in identifying efficient airports and improving operational integration within the aviation sector.

Ultimately, this study represents one of the pioneering applications of the PFAHP–CoCoSo approach in evaluating the operational performance of airports in Turkey.

3 Methodology

This study is not intended to evaluate the overall efficiency or comprehensive performance of airports; rather, it focuses on analyzing the level of operational activities. Airports are ranked according to their operational performance based on a predefined set of criteria, beginning with the highest performance score.

The methodological framework of the research is presented in Figure 1. The process begins with evaluations obtained from decision-makers and concludes with the validation of the ranking results for the alternatives.

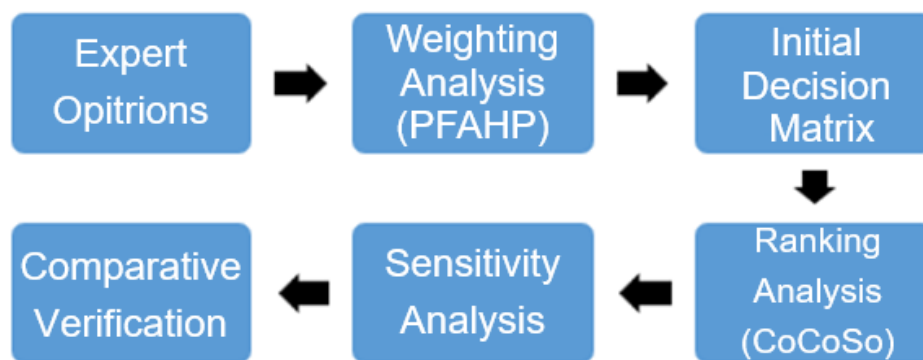


Figure 1. The methodological structure of the research

Note: PFAHP—Pythagorean fuzzy analytic hierarchy process.

To evaluate and rank airports within the assessment process, this study proposes a novel methodological framework integrating the PFAHP and CoCoSo methods. The PFAHP technique was employed to determine the criterion weights because it enables more effective handling of uncertainty and ambiguity in experts' linguistic evaluations compared to traditional fuzzy approaches. In addition, PFAHP facilitates the aggregation of multiple expert judgments and supports consensus-based decision-making.

The CoCoSo method was adopted for ranking the alternatives due to its robust and compromise-oriented evaluation mechanism. By integrating the SAW and Weighted Product Model (WPM) approaches, CoCoSo combines the interpretability of additive aggregation with the sensitivity of multiplicative models within a unified framework. This integration enhances the consistency and stability of the ranking results.

Accordingly, the combined use of PFAHP and CoCoSo provides a comprehensive and systematic framework for evaluating airport operations. The proposed approach establishes a logical basis for determining performance priorities and final alternative scores while focusing on the optimal compromise between ideal and non-ideal solutions. Furthermore, CoCoSo employs three distinct aggregation strategies to calculate performance scores, thereby improving the robustness of the evaluation process.

Expert judgments were obtained through structured face-to-face interviews. The participating experts evaluated and ranked 11 criteria based on their perceived importance. The collected data were subsequently processed and analyzed using Microsoft Excel. During the evaluation process, several computational matrices and comparative assessment tables were generated.

Given the large number of alternatives and extensive computational procedures involved in the analysis, it was not feasible to include all calculation tables in the manuscript. The methodological framework and implementation steps of the proposed approach are presented in the following sections.

3.1 Pythagorean Fuzzy Analytic Hierarchy Process Method

PFAHP is considered an effective approach for addressing subjective, uncertain, and complex decision-making problems. The method was developed to model uncertainty and vagueness in human judgments during MCDM

processes. Unlike the classical AHP method, PFAHP employs linguistic variables and Pythagorean fuzzy numbers rather than precise numerical evaluations. In this context, linguistic expressions such as “important” and “very important” are transformed into fuzzy representations to better reflect expert judgments under uncertainty.

In the present study, the geometric mean of the evaluations obtained from four experts was calculated to aggregate individual judgments into a unified decision matrix. This aggregation process facilitates the integration of diverse expert opinions and supports consensus-based evaluation.

The implementation steps of the PFAHP method are presented below.

Step 1: Utilizing the linguistic scale introduced by Ilbahar et al. [32] (see Table 3), the experts’ assessments are consolidated to construct the aggregated pairwise comparison matrix $A = (a_{ik})_{(mm)}$. Table 3 illustrates the linguistic terms and their corresponding Pythagorean fuzzy representations.

Table 3. Linguistic terms for importance weights of criteria

Linguistic Variables	Pythagorean Fuzzy Numbers			
	μ_L	μ_U	v_L	v_U
Certainly low importance (CLI)	0.00	0.00	0.90	1.00
Very low importance (VLI)	0.10	0.20	0.80	0.90
Low importance (LI)	0.20	0.35	0.65	0.80
Below average importance (BAI)	0.35	0.45	0.55	0.65
Average importance (AI)	0.45	0.55	0.45	0.55
Above average importance (AAI)	0.55	0.65	0.35	0.45
High importance (HI)	0.65	0.80	0.20	0.35
Very high importance (VHI)	0.80	0.90	0.10	0.20
Certainly high importance (CHI)	0.90	1.00	0.00	0.00
Exactly equal (EE)	0.1965	0.1965	0.1965	0.1965

The linguistic evaluation scales and their corresponding numerical representations were adopted from the conventional FAHP methodology without any modification.

Step 2: Eqs. (1) and (2) are employed to construct the difference matrices ($D = (d_{ik})_{(m \times m)}$), which capture the lower and upper boundary values of the membership and non-membership functions.

$$d_{ikL} = \mu_{ikL}^2 - v_{ikU}^2 \quad (1)$$

$$d_{ikU} = \mu_{ikU}^2 - v_{ikL}^2 \quad (2)$$

Step 3: Interval multiplicative matrix $S = (s_{ik})_{(m \times m)}$ is computed using Eq. (3) and (4):

$$s_{ikL} = \sqrt{1000^{d_{ikL}}} \quad (3)$$

$$s_{ikU} = \sqrt{1000^{d_{ikU}}} \quad (4)$$

Step 4: The determinacy value $\tau = (\tau_{ik})_{(m \times m)}$ is calculated using Eq. (5):

$$\tau_{ik} = 1 - (\mu_{ikU}^2 - \mu_{ikL}^2) - (v_{ikU}^2 - v_{ikL}^2) \quad (5)$$

Step 5: Eq. (6) is applied to derive the weight matrix $T = (t_{ik})_{(m \times m)}$ by incorporating the determinacy degrees into the matrix $S = (s_{ik})_{(m \times m)}$ before carrying out the normalization process.

$$t_{ik} = \left(\frac{s_{ikL} + s_{ikU}}{2} \right) \tau_{ik} \quad (6)$$

Step 6: To derive the final normalized weights of the criteria, the priority values (w_i) are adjusted in accordance with Eq. (7).

$$w_i = \frac{\sum_{k=1}^m t_{ik}}{\sum_{i=1}^m \sum_{k=1}^m t_{ik}} \quad (7)$$

3.2 Combined Compromise Solution Method

The CoCoSo method is an MCDM approach developed to evaluate and rank alternatives by considering the relationships between decision alternatives and ideal solution values. The method integrates compromise-based evaluation principles to generate stable and consistent ranking results. Following the identification of the alternatives and evaluation criteria, the computational procedure is implemented through the following steps [29].

Step 1: The initial decision matrix is formulated as shown in Eq. (8).

$$X_{ij} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} \quad (8)$$

where, $i = 1, 2, \dots, m$ (i : alternative); $j = 1, 2, \dots, n$ (j : criterion); X_{ij} : value of alternative i for criterion j .

Step 2: The decision matrix is normalized for benefit and cost separately. Eq. (9) is used for the benefit criterion.

$$r_{ij} = \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}} \quad (9)$$

where, r_{ij} : normalized value of alternative i for criterion j .

Eq. (10) is used for the cost criterion.

$$r_{ij} = \frac{x_j^{\max} - x_{ij}}{x_j^{\max} - x_j^{\min}} \quad (10)$$

Step 3: To evaluate each alternative, the cumulative weighted comparability sequence and the overall power-weighted comparability sequence were derived, resulting in the measures S_i and P_i presented in Equations (11) and (12).

$$S_i = \sum_{j=1}^n (w_j r_{ij}) \quad (11)$$

where, S_i indicates the total weighted comparability sequence for each alternative i . The value is obtained according to the gray relational generation approach.

$$P_i = \sum_{j=1}^n (r_{ij})^{w_j} \quad (12)$$

where, w_j indicates the weight of each criterion j and P_i indicates the total power weight of the comparability sequence for each alternative i .

Step 4: The relative performance weights of the alternatives are computed using three distinct aggregation strategies. The mathematical formulations of these strategies are provided in Eqs. (13)–(15).

Eq. (13) employs an additive aggregation mechanism based on the multiplication of normalized performance values and criterion weights. This formulation measures the overall utility of the alternatives while linearly incorporating criterion importance into the evaluation process. Therefore, unlike uniform weighting approaches, the method enables differentiated contributions of the criteria to the final performance scores.

$$k_{ia} = \frac{P_i + S_i}{\sum_{i=1}^m (P_i + S_i)} \quad (13)$$

where, k_{ia} , k_{ib} and k_{ic} indicate aggregation strategy a , b , and c for alternative i .

Eq. (14) combines the additive aggregation and power-based transformation of the weighted comparability sequences. Through the evaluation of the alternatives' proximity to the ideal solution, the equation enables a more balanced and stable determination of the relative performance priorities of the alternatives.

$$k_{ib} = \frac{S_i}{i^{\min} S_i} + \frac{P_i}{i^{\min} P_i} \quad (14)$$

Eq. (15) employs a multiplicative aggregation mechanism in which the normalized evaluation values are weighted through exponential criterion coefficients. Owing to the sensitivity of multiplicative models, this formulation

effectively captures the impact of variations in individual criterion performance. As a result, significant deviations in a single criterion are more explicitly reflected in the overall performance assessment.

$$k_{ic} = \frac{\lambda (S_i) + (1 - \lambda) (P_i)}{(\lambda)^{i^{\max}} (S_i) + (1 - \lambda)^{i^{\max}} (P_i)} \quad (15)$$

The value of λ is chosen by the decision makers, and the value of λ is generally preferred as 0.5.

Step 5: The final ranking of the alternatives is calculated as shown in Eq. (16).

$$k_i = (k_{ia}k_{ib}k_{ic})^{\frac{1}{3}} + \frac{1}{3} (k_{ia} + k_{ib} + k_{ic}) \quad (16)$$

The highest value obtained represents the best alternative according to the CoCoSo method.

4 Case Study and Result

In this section of the study, the operational performances of airports in Turkey are evaluated using the integrated PFAHP–CoCoSo methodology based on data obtained from the DHMİ. All data used in the CoCoSo analysis were collected as secondary data from DHMİ sources, whereas the data employed in the PFAHP analysis were obtained through expert evaluations.

The dataset initially included 56 airport alternatives, as presented in Table 4. However, four airports were excluded from the analysis because they are not actively operated for passenger transportation, are only used for limited special-purpose operations, and lack sufficient statistical data. Consequently, the empirical application of the study covers 52 active airports.

Table 4. Information on alternatives

Code	Alternative	Code	Alternative	Code	Alternative	Code	Alternative	Code	Alternative
A ₁	Atatürk	A ₁₂	Uşak	A ₂₃	Kahramanmaraş	A ₃₄	Erzurum	A ₄₅	Gaziantep
A ₂	Sabiha Gökçen	A ₁₃	Yenişehir	A ₂₄	Kapadokya	A ₃₅	Erzincan	A ₄₆	Adıyaman
A ₃	İstanbul	A ₁₄	Hasan Polatkan	A ₂₅	Kayseri	A ₃₆	Ahmed-i Hani	A ₄₇	Gap
A ₄	Çorlu	A ₁₅	Cengiz Topel	A ₂₆	Nuri Demirağ	A ₃₇	Harakani	A ₄₈	Diyarbakır
A ₅	Koca Seyit	A ₁₆	Esenboğa	A ₂₇	Caycuma	A ₃₈	Şehit Bülent Aydın	A ₄₉	Mardin
A ₆	Çanakkale	A ₁₇	Konya	A ₂₈	Kastamonu	A ₃₉	Malatya	A ₅₀	Batman
A ₇	Adnan	A ₁₈	Antalya	A ₂₉	Sinop	A ₄₀	Elazığ	A ₅₁	Şerafettin
A ₈	Çardak	A ₁₉	Gazipasa	A ₃₀	Çarşamba	A ₄₁	Bingöl	A ₅₂	Siirt
A ₉	Dalaman	A ₂₀	Süleyman	A ₃₁	Merzifon	A ₄₂	Ferit Melen		
A ₁₀	Milas-Bodrum	A ₂₁	Adana	A ₃₂	Trabzon	A ₄₃	Sultan Alparslan		
A ₁₁	Zafer	A ₂₂	Hatay	A ₃₃	Ordu-Giresun	A ₄₄	Yüksekova-S. Eyyubi		

Previous studies on airport operational performance evaluation were reviewed to identify the relevant assessment criteria, and a preliminary criteria list was established. Subsequently, the availability of data for these criteria was examined using DHMİ data sources. As a result, eleven criteria with accessible and reliable data were selected for the analysis. The operational performance of the airports is evaluated based on the 11 criteria presented in Table 5. These criteria also address Research Question 1 (RQ1).

The selected criteria were determined through an extensive review of the airport performance evaluation literature and expert consultations. The expert panel consisted of professionals with substantial experience in aviation operations, including a ground services operations manager at Antalya Airport with 18 years of experience, a business development manager at Sabiha Gökçen International Airport with 20 years of experience, a technical coordinator at Istanbul Airport with 25 years of experience, and a technical manager at Esenboğa Airport with 14 years of experience.

The criterion values of the alternatives were obtained from 2019 data. To ensure data reliability and avoid the distortions caused by the COVID-19 pandemic, the analysis was based on data from the last operational year before the pandemic. Since airport operations in Turkey are influenced by fluctuations in tourism demand, trade activities, and passenger mobility, performance outcomes may vary across different years. Therefore, analyses conducted using alternative datasets or different time periods may produce variations in the ranking results.

Table 5. The evaluation criteria for operational performance analysis of airports

Code	Criterion	Criterion Type	Code	Criterion	Criterion Type	Code	Criterion	Criterion Type
C ₁	Aircraft traffic domestic line	Benefit	C ₅	Carried load (ton), domestic line	Benefit	C ₉	VIP-CIP areas	Benefit
C ₂	Aircraft traffic international line	Benefit	C ₆	Carried load (ton), international line	Benefit	C ₁₀	Passport control point	Benefit
C ₃	Number of passengers domestic line	Benefit	C ₇	Areas serving passengers (m ²), Domestic and international lines	Benefit	C ₁₁	Number of conveyors	Benefit
C ₄	Number of passengers international line	Benefit	C ₈	Passenger terminals total area (m ²), domestic and international lines	Benefit			

Traffic-related indicators such as the number of flights, passenger volume, and cargo transportation capacity constitute the operational activity levels of airports. Together with other service-related indicators, these criteria represent important measures for evaluating airport operational performance. However, operational performance assessment alone may not comprehensively capture broader dimensions such as service quality, operational efficiency, sustainability, and customer satisfaction.

Accordingly, criteria including delays, security performance, revenues, operational costs, environmental impacts, service quality, and customer satisfaction were excluded from the present analysis due to data limitations. These dimensions may be incorporated into future studies through alternative datasets and evaluation frameworks.

All criteria included in the analysis were treated as benefit-oriented (utility) criteria. In other words, higher criterion values were assumed to indicate superior operational performance and greater utility. Therefore, the evaluation framework was constructed based on a utility maximization perspective.

Criterion C₇ (Areas Serving Passengers) represents the actively utilized passenger service areas, whereas Criterion C₈ (Passenger Terminals) refers to the total terminal area. Consequently, these two criteria reflect different operational dimensions and should not be interpreted as identical measures.

Each criterion was evaluated according to the availability and accessibility of the relevant data. Although some airports contained limited or incomplete data for specific criteria, this situation did not significantly affect the comparability of operationally similar airports within the analysis framework. Furthermore, Criterion C₉ was specifically included to provide additional information regarding passenger traffic intensity.

4.1 Application of the Pythagorean Fuzzy Analytic Hierarchy Process Method

In this section, the PFAHP approach was applied to determine the weights of the evaluation criteria. Expert assessments were collected using the linguistic variables presented in Table 3, and the corresponding pairwise comparison matrices are provided in Appendix 2. The linguistic judgments were subsequently converted into Pythagorean fuzzy numbers based on the predefined fuzzy scale. By following the computational procedure described in Section 3.1, the final criterion weights were obtained and are reported in Table 6.

Table 6. Criterion weights from Pythagorean fuzzy analytic hierarchy process (PFAHP)

Criterion	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇	C ₈	C ₉	C ₁₀	C ₁₁
Weight (w_i)	0.1543	0.0978	0.1715	0.1277	0.0934	0.1291	0.0677	0.0390	0.0164	0.0567	0.0464
Ranking	2	5	1	4	6	3	7	10	11	8	9

According to the findings, C₃ (“Number of Domestic Passengers”), C₁ (“Domestic Aircraft Traffic”), and C₆ (“International Cargo Volume”) emerged as the most significant criteria influencing airport operational performance.

To verify the consistency of the criterion weights, the Consistency Ratio (CR) was calculated using the classical AHP approach. The CR value was found to be 0.045, which is below the acceptable threshold of 0.10, indicating that the pairwise comparison matrix is consistent.

According to the PFAHP results, C₃ (“Number of Domestic Passengers”) was identified as the most important criterion, whereas C₁₁ (“Number of Conveyors”) had the lowest priority weight. The weighting values represent the

relative importance of each criterion in achieving the evaluation objective. The comparatively high weight assigned to C_3 indicates the substantial influence of domestic passenger traffic on airport operational performance.

In addition, the C_3 criterion includes complete and consistent data for all evaluated airports, which enhances its contribution to the overall ranking analysis. By contrast, the lower-ranked criteria— C_9 (“Passenger Terminal Total Area”), C_8 (“VIP-CIP Areas”), and C_{11} (“Number of Conveyors”)—were found to have a relatively limited influence on the final CoCoSo ranking results. This may be attributed to the fact that these infrastructure-related criteria are commonly available across most airports due to standardized fixed investments.

4.2 Application of the CoCoSo Method

Initially, the decision matrix for the CoCoSo analysis was formulated in accordance with Eq. (8), and the complete matrix is provided in Appendix 3. The operational performance data of Turkish airports were collected from the 2019 report issued by the DHMİ [45].

The normalization procedure was conducted using the original dataset without manual adjustment or transformation beyond the standard computational process. The CoCoSo calculations were performed in Microsoft Excel based on the formulations presented in Eqs. (9)–(16), and the detailed computational outputs are reported in Appendices 11–13.

Several criteria values in the decision matrix were recorded as zero. These values represent the non-occurrence of the relevant operational activity rather than missing observations. The dataset used in the analysis was obtained from and verified by DHMİ, the official authority responsible for airport statistics in Turkey.

The final CoCoSo results and the corresponding airport performance rankings are presented in Table 7.

Table 7. CoCoSo results

Alternative	k_i	Rank	Alternative	k_i	Rank	Alternative	k_i	Rank	Alternative	k_i	Rank
A ₁	32.2381	4	A ₁₄	1.2473	51	A ₂₇	1.4835	49	A ₄₀	3.6778	20
A ₂	49.5244	2	A ₁₅	1.5538	48	A ₂₈	1.8935	44	A ₄₁	1.6205	47
A ₃	74.8457	1	A ₁₆	25.9353	5	A ₂₉	2.2965	36	A ₄₂	4.8125	14
A ₄	1.9130	43	A ₁₇	4.0525	19	A ₃₀	4.5647	16	A ₄₃	2.6591	33
A ₅	2.4072	35	A ₁₈	35.7670	3	A ₃₁	1.7527	45	A ₄₄	1.6570	46
A ₆	2.0303	41	A ₁₉	3.2162	25	A ₃₂	9.0829	10	A ₄₅	6.3929	12
A ₇	23.5894	6	A ₂₀	2.0936	39	A ₃₃	4.3024	17	A ₄₆	2.7150	30
A ₈	3.6015	21	A ₂₁	11.4614	7	A ₃₄	4.1784	18	A ₄₇	3.5584	22
A ₉	10.2280	8	A ₂₂	4.6859	15	A ₃₅	2.8097	29	A ₄₈	6.2261	13
A ₁₀	9.8232	9	A ₂₃	2.1351	37	A ₃₆	2.5252	34	A ₄₉	3.3846	23
A ₁₁	1.9860	42	A ₂₄	2.7119	31	A ₃₇	2.9104	28	A ₅₀	3.1380	26
A ₁₂	1.4038	50	A ₂₅	6.4083	11	A ₃₈	2.1313	38	A ₅₁	2.0369	40
A ₁₃	2.6604	32	A ₂₆	2.9629	27	A ₃₉	3.3012	24	A ₅₂	0.9770	52

In the CoCoSo method, the alternative with the highest “ k_i ” value is considered the best-performing option [46]. The CoCoSo analysis was conducted using the criterion weights derived from the PFAHP method.

According to the analysis results, A₃ (Istanbul Airport) achieved the highest operational performance score among the evaluated airports. It was followed by A₂ (Sabiha Gökçen International Airport), A₁₈ (Antalya Airport), A₁ (Istanbul Atatürk Airport), and A₁₆ (Esenboğa Airport), respectively. These findings indicate that airports with high passenger and traffic capacities achieved superior operational performance outcomes under the proposed evaluation framework.

By contrast, A₅₂ (Siirt Airport) obtained the lowest “ k_i ” value and was therefore identified as the lowest-performing airport in the analysis.

The complete ranking results were generated according to the calculated “ k_i ” scores, while the top 10 and bottom 10 airports are illustrated in Figure 2.

The C_3 (“Number of Domestic Passengers”) and C_1 (“Domestic Aircraft Traffic”) criteria, which received the highest weights in the PFAHP analysis, had a substantial influence on the CoCoSo ranking results. This can be attributed to the fact that these criteria represent the most fundamental and consistently observed operational activities across all evaluated airports. Consequently, these criteria contributed significantly to the comparability and stability of the evaluation framework.

Since the criterion datasets exhibit internal consistency, the relative influence of each criterion on the alternatives remained proportionate within the analysis. Furthermore, differences in the scale and magnitude of the criterion values did not significantly distort the final ranking outcomes due to the normalization process applied in the CoCoSo methodology.

The CoCoSo results primarily reflect the relative operational performance levels of the airports within the selected evaluation framework rather than representing an absolute measure of overall airport superiority.

In addition, the ranking results obtained solely from the C_3 criterion differed from the final CoCoSo rankings, indicating that the integrated evaluation framework captures the combined influence of multiple operational criteria rather than relying on a single dominant factor.

Finally, separate examinations of the top five and bottom five airports demonstrated that their relative ranking positions remained stable within their respective groups, supporting the robustness and consistency of the obtained results.

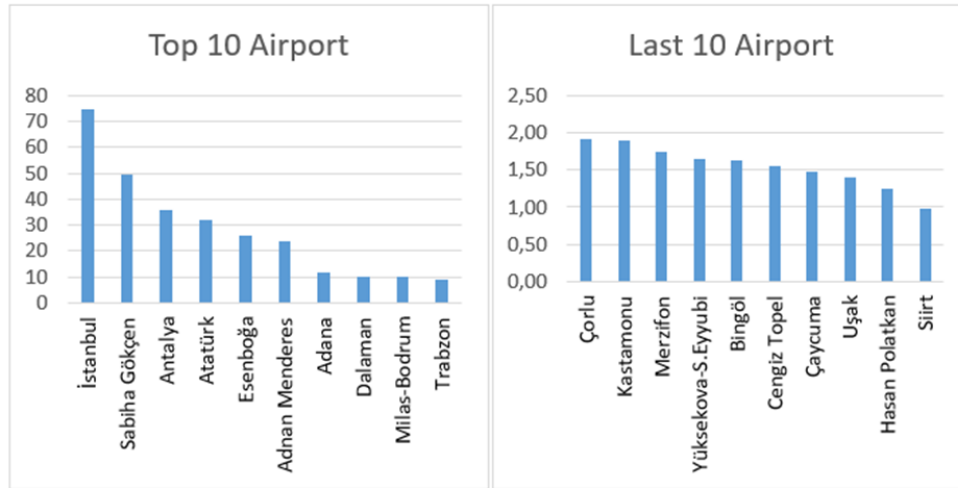


Figure 2. Top 10 and last 10 airports

To assess the robustness of the proposed framework, the $\lambda = 0.5$ parameter employed in the CoCoSo method was analyzed under several sensitivity scenarios. While variations in the λ coefficient influenced the magnitude of the “ k_i ” scores, the ranking positions of the alternatives remained unchanged. This result demonstrates the stability of the proposed ranking structure against parameter variation.

Furthermore, sensitivity analysis was performed by varying the criterion weights in order to evaluate the reliability of the obtained results under different weighting conditions. The detailed findings of the sensitivity analysis are discussed in the subsequent section.

4.3 Sensitivity Analysis and Validation of the Results

This section evaluates the robustness and reliability of the findings obtained in the previous analysis. The validation procedure was implemented in three stages. In the first stage, sensitivity analysis was performed on the $\lambda = 0.5$ parameter employed in the CoCoSo method. In the second stage, the stability of the airport rankings was examined under different criterion weight scenarios. Finally, the performance of the proposed PFAHP–CoCoSo framework was validated through comparisons with alternative fuzzy MCDM approaches.

4.3.1 Effect of changes in $\lambda = 0.5$ on ranking results

To evaluate the robustness of the proposed CoCoSo framework, sensitivity analysis was conducted using alternative parameter settings of $\lambda = 0.3$ and $\lambda = 0.7$ instead of the baseline value of $\lambda = 0.5$. The results showed that, although the numerical values of the “ k_i ” coefficients changed under different parameter scenarios, the relative ranking positions of the alternatives remained stable. This demonstrates the robustness and consistency of the obtained ranking structure with respect to variations in the λ coefficient.

4.3.2 Effect of changes in criteria weights on ranking results

Given that the evaluations provided by decision-makers determine the criterion weights, the effect of variations in these evaluations on the ranking outcomes was assessed through sensitivity analysis. In this context, several experimental scenarios were developed using the framework presented in Table 8 [47] to systematically adjust the criterion weights.

In the first stage of the sensitivity analysis, thirteen different scenarios were constructed to examine how variations in criterion weights affect the ranking results. The first scenario corresponds to the baseline ranking obtained from the original model. The second scenario assumes equal importance for all criteria. In the remaining scenarios, the weight of one criterion is kept constant while the other criteria are assigned equal weights. Based on the number of criteria considered in this study, a total of thirteen scenarios were generated (Appendix 1).

Table 8. CoCoSo results

Scenarios	Explanation
Scenario 1	Current
Scenario 2	Equal weights ($w_{C_1} = w_{C_2} = \dots = w_{C_{11}} = 0.0909$)
Scenario 3	C_1 's weight is the same (0.1543), the rest are the same (0.0846)
Scenario 3	C_2 's weight is the same (0.0978), the rest are the same (0.0902)
Scenario 5	C_3 's weight is the same (0.1715), the rest are the same (0.0829)
Scenario 6	C_4 's weight is the same (0.1277), the rest are the same (0.0872)
Scenario 7	C_5 's weight is the same (0.0934), the rest are the same (0.0907)
Scenario 8	C_6 's weight is the same (0.1291), the rest are the same (0.0871)
Scenario 9	C_7 's weight is the same (0.0677), the rest are the same (0.0932)
Scenario 10	C_8 's weight is the same (0.0390), the rest are the same (0.0961)
Scenario 11	C_9 's weight is the same (0.0164), the rest are the same (0.0984)
Scenario 12	C_{10} 's weight is the same (0.0567), the rest are the same (0.0943)
Scenario 13	C_{11} 's weight is the same (0.0464), the rest are the same (0.0954)

As shown in Appendix 1, alternatives A_2 , A_3 , A_{12} , A_{14} , A_{32} , and A_{52} maintain stable ranking positions across all scenarios. In particular, A_3 consistently ranks first, A_2 consistently ranks second, and A_{52} consistently ranks last. For the remaining alternatives, minor positional changes are observed across scenarios.

According to the CoCoSo results, A_1 ranked 3rd in 10 out of 13 scenarios, A_{18} ranked 4th in 10 out of 13 scenarios, and A_7 ranked 5th in 11 out of 13 scenarios. Similarly, A_{27} and A_{41} consistently occupied the 48th and 49th positions in 12 out of 13 scenarios, respectively. While the top-ranked and bottom-ranked alternatives remain largely stable, intermediate-ranked airports exhibit minor variations in their relative positions.

The ranking stability is further illustrated in Figure 3. As can be observed, the results obtained from the proposed model show high consistency across different experimental scenarios, indicating that the ranking structure is generally robust to changes in criterion weights.

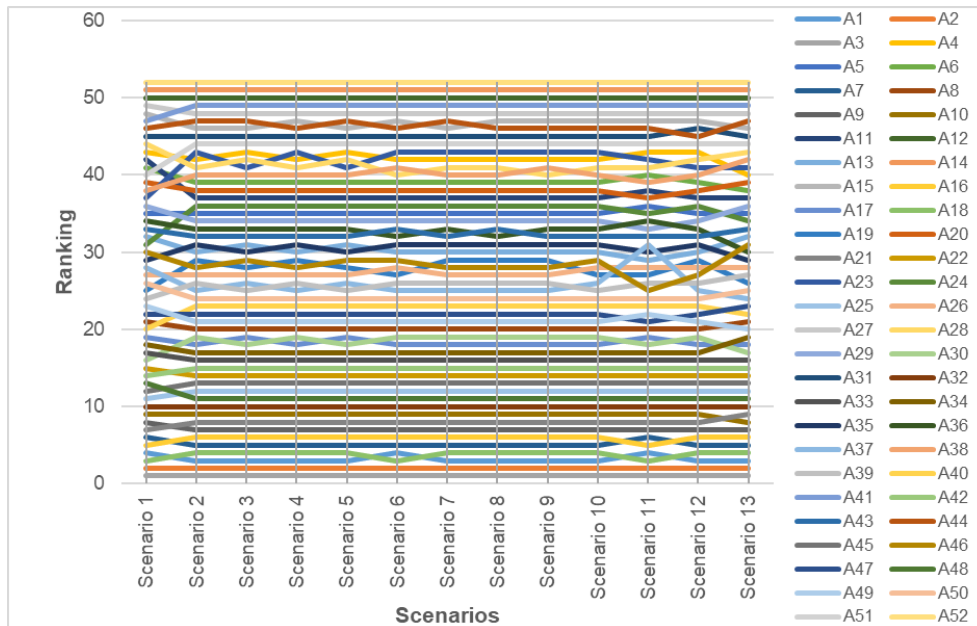


Figure 3. Changes in ranking under different weighting scenarios

According to the scenario-based sensitivity results, alternatives A_{19} , A_{23} , A_{28} , A_{30} , A_{37} , A_{39} , and A_{46} are found to be more sensitive to variations in criterion weights than other airports. Conversely, airports with lower overall performance levels tend to exhibit greater ranking stability under changing weight structures.

Furthermore, an additional robustness check was performed by excluding the most influential criterion (C_3). The results show that the top two and bottom three ranked airports remain stable, while the remaining alternatives experience only marginal changes in their ranking positions, limited to a single-rank deviation.

4.3.3 Comparative analysis with other multi-criteria decision-making methods

To compare the results obtained from the proposed PFAHP–CoCoSo framework, two additional MCDM methods, namely MAIRCA [48] and MABAC [49], were employed. The comparative results are presented in Table 9.

The results indicate that the MAIRCA and MABAC methods produce identical ranking outcomes. In the CoCoSo results, the ranking of the top two alternatives is consistent with those obtained from the other methods.

The Spearman's rank correlation coefficient (SRCC) was calculated to assess the consistency among the ranking results of the three methods. The correlation between CoCoSo and both MAIRCA and MABAC was found to be 0.9968, while the correlation between MAIRCA and MABAC was 1.00. All correlation values exceed 0.80, indicating a high level of agreement among the methods. These findings suggest a strong consistency across the applied MCDM approaches and support the reliability of the proposed PFAHP–CoCoSo framework.

Table 9. Comparison of results with other MCDM methods

CoCoSo			MAIRCA			MABAC			CoCoSo			MAIRCA			MABAC		
	k_i	Rank		U_i	Rank		S_i	Rank		k_i	Rank		U_i	Rank		S_i	Rank
A ₁	32.2381	4		0.01234	4		0.291	4	A ₂₇	1.48	49		0.01913	49		-0.061	49
A ₂	49.5244	2		0.00836	2		0.498	2	A ₂₈	1.89	44		0.01905	44		-0.057	44
A ₃	74.8457	1		0.00264	1		0.796	1	A ₂₉	2.29	36		0.01896	36		-0.052	36
A ₄	1.9130	43		0.01905	45		-0.057	45	A ₃₀	4.56	16		0.01852	16		-0.029	16
A ₅	2.4072	35		0.01893	35		-0.051	35	A ₃₁	1.75	45		0.01908	47		-0.059	47
A ₆	2.0303	41		0.01903	43		-0.056	43	A ₃₂	9.08	10		0.01753	10		0.021	10
A ₇	23.5894	6		0.01429	6		0.190	6	A ₃₃	4.30	17		0.01856	17		-0.032	17
A ₈	3.6015	21		0.01872	22		-0.040	22	A ₃₄	4.17	18		0.01858	18		-0.033	18
A ₉	10.2280	8		0.01729	8		0.034	8	A ₃₅	2.80	29		0.01887	29		-0.047	29
A ₁₀	9.8232	9		0.01738	9		0.029	9	A ₃₆	2.52	34		0.01890	33		-0.049	33
A ₁₁	1.9860	42		0.01902	40		-0.055	40	A ₃₇	2.91	28		0.01881	27		-0.045	27
A ₁₂	1.4038	50		0.01914	50		-0.061	50	A ₃₈	2.13	38		0.01898	37		-0.053	37
A ₁₃	2.6604	32		0.01891	34		-0.049	34	A ₃₉	3.30	24		0.01877	25		-0.042	25
A ₁₄	1.2473	51		0.01915	52		-0.062	52	A ₄₀	3.67	20		0.01870	20		-0.039	20
A ₁₅	1.5538	48		0.01912	48		-0.060	48	A ₄₁	1.62	47		0.01908	46		-0.058	46
A ₁₆	25.9353	5		0.01376	5		0.217	5	A ₄₂	4.81	14		0.01844	14		-0.025	14
A ₁₇	4.0525	19		0.01863	19		-0.035	19	A ₄₃	2.65	33		0.01890	32		-0.049	32
A ₁₈	35.7670	3		0.01154	3		0.333	3	A ₄₄	1.65	46		0.01902	41		-0.055	41
A ₁₉	3.2162	25		0.01875	24		-0.041	24	A ₄₅	6.39	12		0.01812	12		-0.009	12
A ₂₀	2.0936	39		0.01902	42		-0.055	42	A ₄₆	2.71	30		0.01888	30		-0.048	30
A ₂₁	11.4614	7		0.01700	7		0.049	7	A ₄₇	3.55	22		0.01872	21		-0.040	21
A ₂₂	4.6859	15		0.01849	15		-0.028	15	A ₄₈	6.22	13		0.01815	13		-0.010	13
A ₂₃	2.1351	37		0.01900	39		-0.054	39	A ₄₉	3.38	23		0.01875	23		-0.041	23
A ₂₄	2.7119	31		0.01889	31		-0.048	31	A ₅₀	3.13	26		0.01880	26		-0.044	26
A ₂₅	6.4083	11		0.01812	11		-0.008	11	A ₅₁	2.03	40		0.01900	38		-0.054	38
A ₂₆	2.9629	27		0.01884	28		-0.046	28	A ₅₂	0.97	52		0.01915	51		-0.062	51

Note: MCDM—multi-criteria decision-making; CoCoSo—combined compromise solution; MABAC—multi-attributive border approximation area comparison; MAIRCA—multi-attributive ideal-real comparative analysis.

The correlation value of 0.9968 between the proposed model and the benchmark methods indicates that minor differences in ranking positions do not materially affect the overall ordering of the alternatives.

According to the results, the outcomes obtained using the proposed CoCoSo method are consistent with those derived from the MAIRCA and MABAC methods.

The MABAC method provides a systematic framework for analyzing and ranking decision alternatives in environments characterized by multiple conflicting evaluation criteria. The method identifies the best alternative by measuring the distance between each alternative's performance values and a predefined border approximation area constructed between ideal and non-ideal solutions [50]. This border area serves as a reference boundary for the evaluation process.

Due to its distance-based structure, MABAC produces stable and consistent ranking results and eliminates the need for pairwise comparisons, thereby reducing computational complexity. The method evaluates alternatives based on their deviations from the border approximation area, where smaller deviations indicate better performance.

The method operates by calculating the distances between ideal and actual performance values and identifies the alternative with the minimum deviation as the best option. It also allows the incorporation of decision-makers' preference information through criterion weighting.

By quantifying the gaps between alternatives in a structured manner, the method provides a coherent basis for performance evaluation. Due to these characteristics, it yields consistent results in both criterion weighting and

overall performance assessment [51]. Accordingly, MAIRCA and MABAC were selected for comparative analysis in this study.

The same normalization procedure was applied across all three methods (CoCoSo, MABAC, and MAIRCA), as presented in Eqs. (9)–(10).

The top 10 alternatives identified by each method are presented in Table 10. The results reveal a high level of consistency across the methods, with the same airports consistently ranked in the 1st, 2nd, and 10th positions. The remaining alternatives exhibit only minor variations, limited to a one-rank difference.

Table 10. Top 10 alternatives according to the methods

Rank	MAIRCA	MABAC	CoCoSo
1	İstanbul	İstanbul	İstanbul
2	Sabiha Gökçen	Sabiha Gökçen	Sabiha Gökçen
3	Antalya	Antalya	Atatürk
4	Atatürk	Atatürk	Antalya
5	Esenboğa	Esenboğa	Adnan Menderes
6	Adnan Menderes	Adnan Menderes	Esenboğa
7	Adana	Adana	Dalaman
8	Dalaman	Dalaman	Milas-Bodrum
9	Milas-Bodrum	Milas-Bodrum	Adana
10	Trabzon	Trabzon	Trabzon

Note: MABAC—multi-attributive border approximation area comparison; MAIRCA—multi-attributive ideal-real comparative analysis; CoCoSo—combined compromise solution.

In the comparative analysis, the PFAHP-derived weights used in the proposed CoCoSo method were also applied to the MAIRCA and MABAC methods to ensure consistency across the evaluation frameworks.

In the MAIRCA method, alternatives with lower U_i values are considered closer to the ideal solution, and the alternative with the minimum U_i value is identified as the optimal one [52]. Similarly, in the MABAC method, the alternative with the highest S_i value is regarded as the best-performing option [49].

The results indicate that alternative A_3 achieves the highest operational performance across all applied methods. Moreover, the MAIRCA and MABAC methods produce identical ranking results. In the CoCoSo results, the top two alternatives are consistent with those obtained from the other methods. In addition, alternatives A_{32} , A_{26} , A_{35} , A_5 , and A_{29} exhibit identical ranking positions across all models.

For the remaining alternatives, only minor variations are observed, typically limited to a shift of one or two ranks. However, these differences do not significantly affect the overall ranking structure. Accordingly, a high degree of agreement is observed between the proposed method and the benchmark methods.

Overall, the comparative analysis confirms the consistency and robustness of the proposed methodology. The results of the comparison are illustrated in Figure 4.

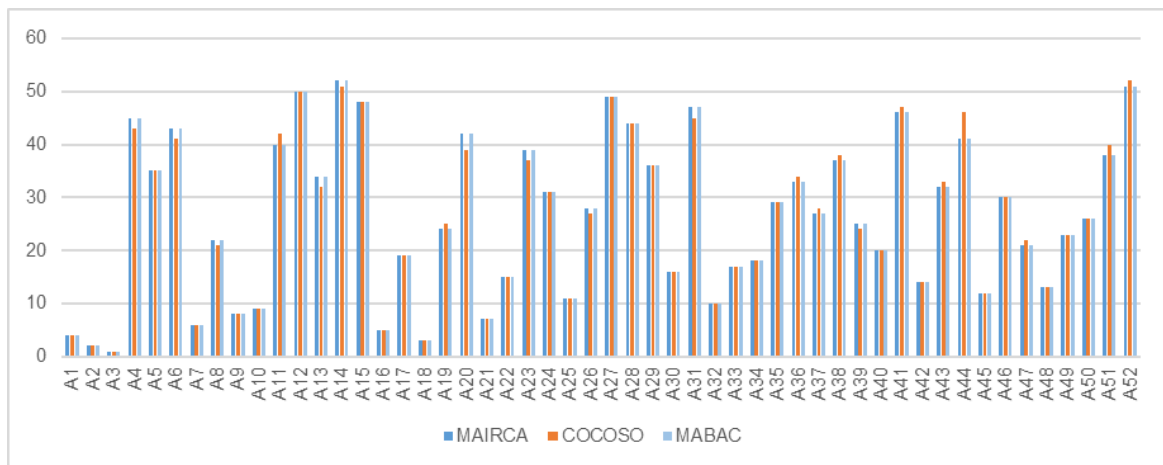


Figure 4. Ranking of the alternatives, the proposed model vs. others

Note: MABAC—multi-attributive border approximation area comparison; MAIRCA—multi-attributive ideal-real comparative analysis; CoCoSo—combined compromise solution.

5 Managerial Implications Limitations

A novel hybrid fuzzy MCDM framework is proposed for the performance evaluation and ranking of airports. Compared with existing fuzzy MCDM approaches, the proposed model provides a more flexible decision-making structure and enables a more comprehensive integration of evaluation criteria within a fuzzy environment. The CoCoSo method contributes to this framework by generating compromise solutions based on simultaneous consideration of ideal and non-ideal reference points.

Based on the empirical findings, the implications and limitations of the study are summarized below.

5.1 Managerial Implications

The findings suggest that improvements in airport operational performance are closely related to higher levels of operational activity, particularly increased air traffic. Higher traffic volumes appear to enhance the interaction among related operational indicators within the evaluation framework.

The proposed model provides a structured decision-support tool that may assist decision-makers in designing investment strategies aligned with airport performance levels and in allocating resources more efficiently. In this context, the identified criterion hierarchy can serve as a reference framework for performance improvement, particularly by strengthening key operational dimensions such as domestic passenger traffic, domestic aircraft movements, and cargo handling.

Furthermore, benchmarking lower-performing airports against higher-performing ones may support the identification and adoption of effective operational practices. According to the PFAHP results, “Number of Domestic Passengers” (C_3) emerges as the most influential criterion, highlighting the importance of domestic passenger demand in airport performance evaluation in Turkey. This suggests that airports with higher domestic traffic levels generally tend to achieve stronger operational performance.

The CoCoSo results indicate that Istanbul Airport achieves the highest operational performance among all alternatives, whereas Siirt Airport is identified as the lowest-performing airport. This finding indicates that smaller regional airports may face challenges in meeting operational performance criteria compared to major hubs, offering implications for targeted policy and resource allocation.

In addition, several airports, including Sabiha Gökçen International Airport, Antalya Airport, Istanbul Atatürk Airport, and Esenboğa Airport, consistently appear among the top-performing alternatives. These results are broadly consistent with Özsoy and Örkücü [53], who also reported strong performance for major Turkish airports. Overall, these airports demonstrate relatively efficient operational structures and strong performance in handling both domestic and international traffic.

Based on 2019 data from the Ministry of Culture and Tourism [54], Turkey, airports located in provinces with higher tourism demand—such as Istanbul, Izmir, Antalya, Muğla, and Trabzon—tend to exhibit stronger operational performance. This pattern is consistent with tourist arrival statistics reported for these provinces (Istanbul: 10 million; Izmir: 3 million; Antalya: 19 million; Muğla: 3 million; Trabzon: 0.5 million).

Regarding criterion importance, after “Number of Domestic Passengers,” the remaining criteria are ranked as domestic aircraft traffic, international cargo volume, total passenger volume, and international aircraft traffic. This ranking may support decision-makers in prioritizing operational improvement strategies.

Finally, the ranking results may also reflect, to some extent, differences in the economic and infrastructural development levels of the cities where airports are located. However, this interpretation should be considered with caution, as airport performance is influenced by multiple operational and contextual factors.

5.2 Limitations

The proposed model was implemented using 2019 data, and no temporal comparison across different periods was conducted. Accordingly, the analysis represents a cross-sectional evaluation.

The model relies exclusively on absolute performance indicators. In addition, four expert evaluations were used to determine the criterion weights, while expert judgments were limited solely to the weighting phase of the analysis.

The evaluation framework is based on eleven operational criteria, and airports are assessed strictly within this set of indicators. Consequently, the performance ranking reflects only operational activities, while dimensions such as safety, delays, costs, revenues, and sustainability were not incorporated into the model. Therefore, the results do not capture the social or strategic roles of airports.

The methodology integrates the PFAHP approach for weight determination and the CoCoSo method for ranking alternatives.

The dataset includes airports located in Turkey only; thus, the findings are not intended for global generalization.

All performance indicators are derived from objective and quantitative data, with no subjective or interpretive variables included in the evaluation process.

The robustness of the results is verified through comparisons with alternative MCDM methods; however, external validation using real-world performance benchmarks was not performed.

Finally, both international and regional airports were evaluated within a unified framework without categorical separation, which may influence the comparability of heterogeneous airport types.

6 Conclusions

This study does not aim to comprehensively evaluate all aspects of airport performance, but rather focuses on operational capacity indicators, including flight traffic, passenger traffic, cargo traffic, and related services.

Airport operational performance analysis has become particularly important in the post-pandemic context. In this study, a PFAHP–CoCoSo integrated MCDM framework is proposed for evaluating the operational performance of airports in Turkey as a real-world decision-making problem. The model incorporates both quantitative performance indicators and expert judgments into the evaluation process.

Due to the inherent subjectivity of human judgment, airport performance evaluation is characterized by uncertainty and ambiguity. Therefore, linguistic assessments cannot always be accurately represented using precise numerical values. In this context, PFSs have been widely adopted in recent studies to handle such uncertainty.

The proposed integrated approach contributes to the MCDM literature by combining PFAHP for criterion weighting and CoCoSo for alternative ranking. The results indicate that the model provides a consistent and structured framework for airport performance evaluation in Turkey. The findings also demonstrate the applicability of fuzzy MCDM techniques in aviation-related decision problems and support the robustness of the proposed model through sensitivity analysis and cross-method validation.

According to expert evaluations, the most influential criterion is “Number of Domestic Passengers” (C_3). The ranking results indicate that the best-performing airports are Istanbul (A_3), Sabiha Gökçen (A_2), Atatürk (A_1), Antalya (A_{18}), Adnan Menderes (A_7), Esenboğa (A_{16}), Dalaman (A_9), Milas-Bodrum (A_{10}), Adana (A_{21}), and Trabzon (A_{32}). In contrast, Siirt (A_{52}) is identified as the lowest-performing airport, followed by Hasan Polatkan (A_{14}) and Uşak (A_{12}).

In the application stage, airport performance was evaluated based on eleven criteria. Criterion weights were determined using the PFAHP method, while the CoCoSo method was applied for ranking alternatives. The outcomes of the sensitivity analysis indicate that the proposed framework maintains its effectiveness and stability despite variations in model parameters and assumptions.

Performance indicators should not be interpreted in isolation, and inter-airport comparisons should account for contextual differences. Understanding the underlying factors affecting performance is essential for meaningful benchmarking and decision-making.

Future research may extend this framework by incorporating multi-year panel data, efficiency-based methods such as DEA or hybrid DEA-MCDM approaches, as well as environmental and service quality indicators. Additionally, clustering airports by size prior to evaluation may provide further analytical depth.

Finally, airport performance evaluation can benefit from the integration of uncertainty-based methods, such as PFSs, which allow decision-makers to model incomplete or inconsistent judgments more effectively. These methods capture uncertainty in a structured way and improve the realism of decision-making models.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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Appendix

Appendix A1. Ranking results of alternatives based on various scenarios

	Scenarios 1	Scenarios 2	Scenarios 3	Scenarios 4	Scenarios 5	Scenarios 6	Scenarios 7	Scenarios 8	Scenarios 9	Scenarios 10	Scenarios 11	Scenarios 12	Scenarios 13
A1	4	3	3	3	3	4	3	3	3	3	4	3	3
A2	2	2	2	2	2	2	2	2	2	2	2	2	2
A3	1	1	1	1	1	1	1	1	1	1	1	1	1
A4	43	42	43	42	43	42	42	42	42	42	43	43	40
A5	35	35	35	35	35	35	35	35	35	35	36	35	35
A6	41	39	39	39	39	39	39	39	39	39	40	39	38
A7	6	5	5	5	5	5	5	5	5	5	6	5	5
A8	21	20	20	20	20	20	20	20	20	20	20	20	21
A9	8	7	7	7	7	7	7	7	7	7	7	7	7
A10	9	9	9	9	9	9	9	9	9	9	9	9	8
A11	42	37	37	37	37	37	37	37	37	37	38	37	37
A12	50	50	50	50	50	50	50	50	50	50	50	50	50
A13	32	30	31	30	31	30	30	30	30	30	29	30	32
A14	51	51	51	51	51	51	51	51	51	51	51	51	51
A15	48	46	46	47	46	47	46	47	47	47	47	47	46
A16	5	6	6	6	6	6	6	6	6	6	5	6	6
A17	19	18	19	18	19	18	18	18	18	18	19	18	18
A18	3	4	4	4	4	3	4	4	4	4	3	4	4
A19	25	29	28	29	28	27	29	29	29	27	27	29	26
A20	39	38	38	38	38	38	38	38	38	38	37	38	39
A21	7	8	8	8	8	8	8	8	8	8	8	8	9
A22	15	14	14	14	14	14	14	14	14	14	14	14	14
A23	37	43	41	43	41	43	43	43	43	43	42	41	41
A24	31	36	36	36	36	36	36	36	36	36	35	36	34
A25	11	12	12	12	12	12	12	12	12	12	12	12	12
A26	27	27	27	27	27	28	27	27	27	28	28	28	28
A27	49	48	48	48	48	48	48	48	48	48	48	48	48
A28	44	41	42	41	42	40	41	41	40	41	41	42	43
A29	36	34	34	34	34	34	34	34	34	34	33	34	36
A30	16	19	18	19	18	19	19	19	19	19	18	19	17
A31	45	45	45	45	45	45	45	45	45	45	45	46	45
A32	10	10	10	10	10	10	10	10	10	10	10	10	10
A33	17	16	16	16	16	16	16	16	16	16	16	16	16
A34	18	17	17	17	17	17	17	17	17	17	17	17	19
A35	29	31	30	31	30	31	31	31	31	31	30	31	29
A36	34	33	33	33	33	32	33	32	33	33	34	33	30
A37	28	25	26	25	26	25	25	25	25	26	31	25	24
A38	38	40	40	40	40	41	40	40	41	40	39	40	42
A39	24	26	25	26	25	26	26	26	26	25	26	26	27
A40	20	23	23	23	23	23	23	23	23	23	23	23	22
A41	47	49	49	49	49	49	49	49	49	49	49	49	49
A42	14	15	15	15	15	15	15	15	15	15	15	15	15
A43	33	32	32	32	32	33	32	33	32	32	32	32	33
A44	46	47	47	46	47	46	47	46	46	46	46	45	47
A45	12	13	13	13	13	13	13	13	13	13	13	13	13
A46	30	28	29	28	29	29	28	28	28	29	25	27	31
A47	22	22	22	22	22	22	22	22	22	22	21	22	23
A48	13	11	11	11	11	11	11	11	11	11	11	11	11
A49	23	21	21	21	21	21	21	21	21	21	22	21	20
A50	26	24	24	24	24	24	24	24	24	24	24	24	25
A51	40	44	44	44	44	44	44	44	44	44	44	44	44
A52	52	52	52	52	52	52	52	52	52	52	52	52	52

Appendix A2. Pairwise comparisons of criteria based on expert's judgements

*	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇	C ₈	C ₉	C ₁₀	C ₁₁
C ₁	EE, EE, EE, EE	HI, AI, BAI, AI	BAI, BAI, BAI, BAI	AI, AI, BAI, AI	AI, AAI, AI, AI	AI, AI, AI, AI	AAI, AI, HI, AAI	AI, HI, AAI, HI	AAI, CHI, CHI, CHI	AAI, AAI, AI, AAI	CHI, AAI, AAI, AAI
C ₂	LI, AI, AAI, AI	EE, EE, EE, EE	VLI, BAI, BAI, BAI	BAI, BAI, BAI, BAI	LI, AI, AI, AI	LI, BAI, AI, AI	BAI, BAI, HI, AI	LI, AAI, AAI, HI	BAI, VHI, CHI, CHI	BAI, AAI, AI, AAI	AAI, AI, AAI, AAI
C ₃	AAI, AAI, AAI, AAI	VHI, AAI, AAI, AAI	EE, EE, EE, EE	AI, AI, BAI, AI	AI, AAI, AI, AI	AI, AI, AI, AI	AAI, AI, HI, AAI	AI, HI, AAI, HI	AAI, CHI, CHI, CHI	AAI, AAI, AAI, AAI	CHI, AAI, AAI, AAI
C ₄	AI, AI, AAI, AI	AAI, AAI, AAI, AAI	AI, AI, AAI, AI	EE, EE, EE, EE	BAI, AI, AI, AI	BAI, BAI, AI, AI	AAI, BAI, HI, AI	BAI, AAI, AAI, HI	AI, VHI, CHI, CHI	AI, AAI, AAI, AAI	VHI, AI, AAI, AAI
C ₅	AI, BAI, AI, AI	HI, AI, AI, AI	AI, BAI, AI, AI	AAI, AI, AI, AI	EE, EE, EE, EE	AI, BAI, BAI, AI	AAI, BAI, AAI, AI	AI, AAI, AAI, AAI	AAI, HI, VHI, VHI	AI, AI, AI, AAI	CHI, AI, AI, AI
C ₆	AI, AI, AI, AI	HI, AAI, AI, AI	AI, AI, AI, AI	AAI, AAI, AI, AI	AI, AAI, AAI, AI	EE, AAI, EE, EE	AAI, AI, AAI, AI	AI, HI, AAI, AAI	AAI, CHI, VHI, VHI	AI, AAI, AI, AAI	CHI, AAI, AI, AI
C ₇	BAI, AI, LI, BAI	AAI, AAI, LI, AI	BAI, AI, LI, BAI	BAI, AAI, LI, AI	BAI, AAI, BAI, AI	BAI, AI, BAI, AI	EE, EE, EE, EE	BAI, HI, BAI, AAI	BAI, CHI, AAI, HI	BAI, AAI, BAI, AI	AAI, AAI, BAI, AI
C ₈	AI, LI, BAI, LI	HI, BAI, BAI, LI	AI, LI, BAI, BAI, LI	AAI, BAI, BAI, LI	AI, BAI, BAI, BAI	AI, LI, BAI, BAI, BAI	AAI, LI, AAI, BAI	EE, EE, EE, EE	AI, AAI, AAI, AAI	AI, BAI, BAI, BAI	VHI, BAI, BAI, BAI
C ₉	BAI, CLI, CLI, CLI	AAI, VLI, CLI, CLI	BAI, CLI, CLI, CLI	AI, VLI, CLI, CLI	BAI, LI, VLI, VLI	BAI, CLI, VLI, VLI	AAI, CLI, BAI, LI	AI, BAI, BAI, BAI	EE, EE, EE, EE	BAI, LI, LI, BAI	HI, LI, LI, LI
C ₁₀	BAI, BAI, AI, BAI	AAI, BAI, AI, BAI	BAI, BAI, BAI, BAI	AI, BAI, BAI, BAI	AI, AI, AI, BAI	AI, BAI, AI, BAI	AAI, BAI, AAI, AI	AI, AAI, AAI, AAI	AAI, AAI, HI, AAI	EE, EE, EE, EE	HI, BAI, AI, BAI
C ₁₁	CLI, BAI, BAI, BAI	BAI, AI, BAI, BAI	CLI, BAI, BAI, BAI	VLI, AI, BAI, BAI	CLI, AI, AI, AI	CLI, BAI, AI, AI	BAI, BAI, AAI, AI	VLI, AAI, AAI, AAI	LI, HI, HI, HI	LI, AAI, AI, AAI	EE, EE, EE, EE

Note: * Alphabetical data is explained in Table 3. Each statement within the cell represents the opinions of four different experts; EE—exactly equal; HI—high importance; AI—average importance; BAI—below average importance; AAI—above average importance; CHI—certainly high importance; LI—low importance; VLI—very low importance; VHI—very high importance; CLI—certainly low importance.

Appendix A3. Initial decision matrix

	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇	C ₈	C ₉	C ₁₀	C ₁₁
A ₁	27,773	92,270	4,236,203	11,876,601	43,647	1,068,441	135,634	387,993	16,065	100	19
A ₂	132,098	97,820	21,505,088	14,055,522	149,979	239,940	115,858	233,461	7,879	70	0
A ₃	80,644	245,763	12,574,641	39,434,579	140,581	1,352,327	698,662	1,464,588	34,010	228	28
A ₄	448	57	71,137	1,816	454	783	2,145	6,521	160	10	3
A ₅	2,318	34	349,883	4,498	2,472	98	9,250	23,240	1,000	0	4
A ₆	935	27	127,080	4,160	914	67	6,544	11,400	360	7	3
A ₇	54,056	22,521	9,031,924	3,333,332	73,172	60,279	87,550	310,978	13,053	37	11
A ₈	3,730	642	529,091	120,691	4,063	2,280	9,260	16,890	749	12	6
A ₉	10,359	18,071	1,583,089	3,321,930	13,528	43,927	69,083	244,406	5,449	66	8
A ₁₀	16,478	11,008	2,464,398	1,873,335	20,359	24,757	46,239	110,613	1,222	36	9
A ₁₁	556	171	57,741	24,285	486	534	0	17,620	746	8	4
A ₁₂	271	8	27,475	508	198	11	780	1,460	42	3	2
A ₁₃	1,791	106	252,269	16,722	2,294	375	5,600	12,716	225	8	5
A ₁₄	6	603	657	88,365	8	1,911	2,000	4,000	250	5	1
A ₁₅	316	34	52,107	5,173	476	113	1,065	2,100	50	5	2
A ₁₆	71,992	18,109	11,463,200	2,277,395	80,578	39,379	88,566	182,000	4,234	41	9
A ₁₇	5,842	874	884,188	124,615	7,114	2,745	12,669	24,175	1,277	12	4
A ₁₈	42,753	154,626	6,958,930	28,720,491	60,708	363,787	61,283	178,637	2,800	116	16
A ₁₉	3,460	3,667	493,485	591,416	4,301	7,113	0	6,700	0	14	4
A ₂₀	624	317	79,866	69,771	623	1,179	2,790	6,770	107	6	4
A ₂₁	25,724	6,183	4,306,031	751,757	34,059	12,702	6,375	12,195	378	13	12
A ₂₂	6,103	2,035	958,106	255,764	8,734	4,226	10,710	43,688	2,992	10	6
A ₂₃	1,912	5	266,136	344	2,195	5	2,270	22,330	35	2	3
A ₂₄	3,091	30	490,125	3,960	5,101	81	2,610	3,500	43	7	3
A ₂₅	12,243	2,308	1,980,252	345,611	19,258	7,362	6,250	22,000	578	13	5
A ₂₆	3,062	31	486,840	4,468	4,022	101	5,765	20,047	794	9	4
A ₂₇	238	76	19,500	9,408	159	163	455	1,430	67	5	2
A ₂₈	505	20	60,835	2,260	428	45	1,565	3,740	152	5	4
A ₂₉	1,036	2	136,862	319	1,105	7	4,147	12,695	810	6	5
A ₃₀	8,389	961	1,347,965	141,368	10,790	3,188	4,725	11,500	720	11	3
A ₃₁	1,097	43	160,842	6,916	1,215	156	450	1,200	18	4	2
A ₃₂	20,032	3,033	3,373,461	397,417	28,257	7,280	11,325	23,745	915	8	8
A ₃₃	6,359	179	1,034,750	25,515	7,721	559	7,750	20,250	1,758	8	6
A ₃₄	5,930	75	984,037	6,650	7,742	153	5,750	12,950	505	12	6
A ₃₅	2,666	8	417,497	958	3,330	20	7,575	27,132	255	9	4
A ₃₆	2,045	0	321,758	0	2,854	0	7,985	23,676	1,394	8	4
A ₃₇	3,192	1	524,462	194	4,751	2	0	35,946	1,975	9	4
A ₃₈	1,620	10	260,174	1,081	2,682	11	0	3,460	50	3	4
A ₃₉	4,581	73	733,026	10,144	6,114	247	2,595	9,545	503	6	4
A ₄₀	5,259	276	861,357	41,145	7,272	1,028	4,955	16,397	540	8	4
A ₄₁	1,389	4	202,753	560	1,757	8	1,450	3,600	58	0	2
A ₄₂	8,197	53	1,406,099	7,268	11,947	97	5,250	14,800	669	6	6
A ₄₃	2,336	12	384,011	1,575	3,449	39	5,888	10,300	240	1	5
A ₄₄	1,195	0	180,506	0	1,886	0	0	6,700	142	0	4
A ₄₅	13,418	2,537	2,152,284	372,092	17,619	6,637	9,433	22,790	637	10	4
A ₄₆	1,584	13	243,207	1,369	2,025	32	4,583	23,011	96	8	6
A ₄₇	4,564	78	715,853	14,094	5,271	369	5,955	12,000	970	4	6
A ₄₈	10,264	592	1,706,191	76,045	12,612	1,738	34,315	86,571	240	12	8
A ₄₉	3,426	17	565,856	2,388	5,058	60	9,042	33,150	1,755	10	5
A ₅₀	3,113	7	522,761	1,249	4,683	37	4,716	20,741	1,119	8	5
A ₅₁	2,281	0	365,865	0	3,364	0	2,347	4,000	381	3	2
A ₅₂	279	0	34,718	0	354	0	576	598	80	0	2

Appendix A4. Normalization for CoCoSo

	k_1	k_2	k_3	k_4	k_5	k_6	k_7	k_8	k_9	k_{10}	k_{11}
A ₁	0.2102	0.3754	0.1970	0.3012	0.2910	0.7901	0.1941	0.2646	0.4724	0.4386	0.6786
A ₂	1.0000	0.3980	1.0000	0.3564	1.0000	0.1774	0.1658	0.1591	0.2317	0.3070	0.0000
A ₃	0.6105	1.0000	0.5847	1.0000	0.9373	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
A ₄	0.0033	0.0002	0.0033	0.0000	0.0030	0.0006	0.0031	0.0040	0.0047	0.0439	0.1071
A ₅	0.0175	0.0001	0.0162	0.0001	0.0164	0.0001	0.0132	0.0155	0.0294	0.0000	0.1429
A ₆	0.0070	0.0001	0.0059	0.0001	0.0060	0.0000	0.0094	0.0074	0.0106	0.0307	0.1071
A ₇	0.4092	0.0916	0.4200	0.0845	0.4879	0.0446	0.1253	0.2120	0.3838	0.1623	0.3929
A ₈	0.0282	0.0026	0.0246	0.0031	0.0270	0.0017	0.0133	0.0111	0.0220	0.0526	0.2143
A ₉	0.0784	0.0735	0.0736	0.0842	0.0902	0.0325	0.0989	0.1665	0.1602	0.2895	0.2857
A ₁₀	0.1247	0.0448	0.1146	0.0475	0.1357	0.0183	0.0662	0.0751	0.0359	0.1579	0.3214
A ₁₁	0.0042	0.0007	0.0027	0.0006	0.0032	0.0004	0.0000	0.0116	0.0219	0.0351	0.1429
A ₁₂	0.0020	0.0000	0.0012	0.0000	0.0013	0.0000	0.0011	0.0006	0.0012	0.0132	0.0714
A ₁₃	0.0135	0.0004	0.0117	0.0004	0.0152	0.0003	0.0080	0.0083	0.0066	0.0351	0.1786
A ₁₄	0.0000	0.0025	0.0000	0.0022	0.0000	0.0014	0.0029	0.0023	0.0074	0.0219	0.0357
A ₁₅	0.0023	0.0001	0.0024	0.0001	0.0031	0.0001	0.0015	0.0010	0.0015	0.0219	0.0714
A ₁₆	0.5450	0.0737	0.5330	0.0578	0.5372	0.0291	0.1268	0.1239	0.1245	0.1798	0.3214
A ₁₇	0.0442	0.0036	0.0411	0.0032	0.0474	0.0020	0.0181	0.0161	0.0375	0.0526	0.1429
A ₁₈	0.3236	0.6292	0.3236	0.7283	0.4047	0.2690	0.0877	0.1216	0.0823	0.5088	0.5714
A ₁₉	0.0261	0.0149	0.0229	0.0150	0.0286	0.0053	0.0000	0.0042	0.0000	0.0614	0.1429
A ₂₀	0.0047	0.0013	0.0037	0.0018	0.0041	0.0009	0.0040	0.0042	0.0031	0.0263	0.1429
A ₂₁	0.1947	0.0252	0.2002	0.0191	0.2271	0.0094	0.0091	0.0079	0.0111	0.0570	0.4286
A ₂₂	0.0462	0.0083	0.0445	0.0065	0.0582	0.0031	0.0153	0.0294	0.0880	0.0439	0.2143
A ₂₃	0.0144	0.0000	0.0123	0.0000	0.0146	0.0000	0.0032	0.0148	0.0010	0.0088	0.1071
A ₂₄	0.0234	0.0001	0.0228	0.0001	0.0340	0.0001	0.0037	0.0020	0.0013	0.0307	0.1071
A ₂₅	0.0926	0.0094	0.0921	0.0088	0.1284	0.0054	0.0089	0.0146	0.0170	0.0570	0.1786
A ₂₆	0.0231	0.0001	0.0226	0.0001	0.0268	0.0001	0.0083	0.0133	0.0233	0.0395	0.1429
A ₂₇	0.0018	0.0003	0.0009	0.0002	0.0010	0.0001	0.0007	0.0006	0.0020	0.0219	0.0714
A ₂₈	0.0038	0.0001	0.0028	0.0001	0.0028	0.0000	0.0022	0.0021	0.0045	0.0219	0.1429
A ₂₉	0.0078	0.0000	0.0063	0.0000	0.0073	0.0000	0.0059	0.0083	0.0238	0.0263	0.1786
A ₃₀	0.0635	0.0039	0.0627	0.0036	0.0719	0.0024	0.0068	0.0074	0.0212	0.0482	0.1071
A ₃₁	0.0083	0.0002	0.0074	0.0002	0.0080	0.0001	0.0006	0.0004	0.0005	0.0175	0.0714
A ₃₂	0.1516	0.0123	0.1568	0.0101	0.1884	0.0054	0.0162	0.0158	0.0269	0.0351	0.2857
A ₃₃	0.0481	0.0007	0.0481	0.0006	0.0514	0.0004	0.0111	0.0134	0.0517	0.0351	0.2143
A ₃₄	0.0448	0.0003	0.0457	0.0002	0.0516	0.0001	0.0082	0.0084	0.0148	0.0526	0.2143
A ₃₅	0.0201	0.0000	0.0194	0.0000	0.0222	0.0000	0.0108	0.0181	0.0075	0.0395	0.1429
A ₃₆	0.0154	0.0000	0.0149	0.0000	0.0190	0.0000	0.0114	0.0158	0.0410	0.0351	0.1429
A ₃₇	0.0241	0.0000	0.0244	0.0000	0.0316	0.0000	0.0000	0.0241	0.0581	0.0395	0.1429
A ₃₈	0.0122	0.0000	0.0121	0.0000	0.0178	0.0000	0.0000	0.0020	0.0015	0.0132	0.1429
A ₃₉	0.0346	0.0003	0.0341	0.0003	0.0407	0.0002	0.0037	0.0061	0.0148	0.0263	0.1429
A ₄₀	0.0398	0.0011	0.0400	0.0010	0.0484	0.0008	0.0071	0.0108	0.0159	0.0351	0.1429
A ₄₁	0.0105	0.0000	0.0094	0.0000	0.0117	0.0000	0.0021	0.0021	0.0017	0.0000	0.0714
A ₄₂	0.0620	0.0002	0.0654	0.0002	0.0796	0.0001	0.0075	0.0097	0.0197	0.0263	0.2143
A ₄₃	0.0176	0.0000	0.0178	0.0000	0.0229	0.0000	0.0084	0.0066	0.0071	0.0044	0.1786
A ₄₄	0.0090	0.0000	0.0084	0.0000	0.0125	0.0000	0.0000	0.0042	0.0042	0.0000	0.1429
A ₄₅	0.1015	0.0103	0.1001	0.0094	0.1174	0.0049	0.0135	0.0152	0.0187	0.0439	0.1429
A ₄₆	0.0119	0.0001	0.0113	0.0000	0.0134	0.0000	0.0066	0.0153	0.0028	0.0351	0.2143
A ₄₇	0.0345	0.0003	0.0333	0.0004	0.0351	0.0003	0.0085	0.0078	0.0285	0.0175	0.2143
A ₄₈	0.0777	0.0024	0.0793	0.0019	0.0840	0.0013	0.0491	0.0587	0.0071	0.0526	0.2857
A ₄₉	0.0259	0.0001	0.0263	0.0001	0.0337	0.0000	0.0129	0.0222	0.0516	0.0439	0.1786
A ₅₀	0.0235	0.0000	0.0243	0.0000	0.0312	0.0000	0.0068	0.0138	0.0329	0.0351	0.1786
A ₅₁	0.0172	0.0000	0.0170	0.0000	0.0224	0.0000	0.0034	0.0023	0.0112	0.0132	0.0714
A ₅₂	0.0021	0.0000	0.0016	0.0000	0.0023	0.0000	0.0008	0.0000	0.0024	0.0000	0.0714

Appendix A5. S_i and P_i for CoCoSo

	S_i	P_i		S_i	P_i		S_i	P_i
A ₁	0.358170347	9.939403361	A ₁₉	0.024968737	6.141354054	A ₃₇	0.021621640	6.065154339
A ₂	0.565185476	9.318211453	A ₂₀	0.010813398	6.920709245	A ₃₈	0.013109875	5.894427428
A ₃	0.862814530	10.83271817	A ₂₁	0.115939863	8.550693542	A ₃₉	0.023932504	7.190913112
A ₄	0.009363550	6.610656726	A ₂₂	0.038302183	7.956185547	A ₄₀	0.027650965	7.492503119
A ₅	0.015674915	6.189033327	A ₂₃	0.011998343	6.523364268	A ₄₁	0.007886741	5.624956261
A ₆	0.010499510	6.704193432	A ₂₄	0.017778831	6.907912102	A ₄₂	0.040920774	7.365308436
A ₇	0.256738704	9.512596215	A ₂₅	0.057792204	8.155691373	A ₄₃	0.017422582	6.779240490
A ₈	0.026581332	7.596907484	A ₂₆	0.020315365	7.070833512	A ₄₄	0.010858864	4.222971799
A ₉	0.100762695	8.892886541	A ₂₇	0.005250159	6.258568634	A ₄₅	0.057577227	8.173725544
A ₁₀	0.096249862	8.789398280	A ₂₈	0.009530003	6.397515846	A ₄₆	0.018074164	6.768586910
A ₁₁	0.011029706	6.139571975	A ₂₉	0.013874874	6.487305932	A ₄₇	0.026713720	7.269831066
A ₁₂	0.004829379	6.001580374	A ₃₀	0.037204002	7.805396891	A ₄₈	0.056064645	7.965781038
A ₁₃	0.016903261	7.066590411	A ₃₁	0.007737983	6.526164835	A ₄₉	0.025037537	7.109193321
A ₁₄	0.004014787	5.489789852	A ₃₂	0.088490116	8.367121968	A ₅₀	0.022531015	6.950583031
A ₁₅	0.005832264	6.365517854	A ₃₃	0.034743526	7.522159480	A ₅₁	0.012225512	5.798124100
A ₁₆	0.284606420	9.468796697	A ₃₄	0.033710463	7.322182918	A ₅₂	0.004217844	3.691764458
A ₁₇	0.031392270	7.774290691	A ₃₅	0.018944112	6.876456803			
A ₁₈	0.399871388	10.00921307	A ₃₆	0.017399717	5.981201685			

Appendix A6. k_i and rank for CoCoSo

	k_{ia}	k_{ib}	k_{ic}	k_i	Rank		k_{ia}	k_{ib}	k_{ic}	k_i	Rank
A ₁	0.0272	91.9051	0.8805	32.2381	4	A ₂₇	0.0165	3.0030	0.5356	1.4835	49
A ₂	0.0261	143.3000	0.8451	49.5244	2	A ₂₈	0.0169	4.1066	0.5478	1.8935	44
A ₃	0.0309	217.8435	1.0000	74.8457	1	A ₂₉	0.0172	5.2132	0.5559	2.2965	36
A ₄	0.0175	4.1229	0.5660	1.9130	43	A ₃₀	0.0207	11.3810	0.6706	4.5647	16
A ₅	0.0164	5.5807	0.5305	2.4072	35	A ₃₁	0.0172	3.6951	0.5587	1.7527	45
A ₆	0.0177	4.4312	0.5741	2.0303	41	A ₃₂	0.0223	24.3075	0.7230	9.0829	10
A ₇	0.0258	66.5250	0.8353	23.5894	6	A ₃₃	0.0199	10.6914	0.6461	4.3024	17
A ₈	0.0201	8.6787	0.6518	3.6015	21	A ₃₄	0.0194	10.3800	0.6289	4.1784	18
A ₉	0.0237	27.5067	0.7690	10.2280	8	A ₃₅	0.0182	6.5812	0.5896	2.8097	29
A ₁₀	0.0235	26.3547	0.7597	9.8232	9	A ₃₆	0.0158	5.9541	0.5129	2.5252	34
A ₁₁	0.0162	4.4103	0.5259	1.9860	42	A ₃₇	0.0161	7.0284	0.5204	2.9104	28
A ₁₂	0.0159	2.8286	0.5136	1.4038	50	A ₃₈	0.0156	4.8620	0.5051	2.1313	38
A ₁₃	0.0187	6.1244	0.6057	2.6604	32	A ₃₉	0.0190	7.9089	0.6169	3.3012	24
A ₁₄	0.0145	2.4870	0.4697	1.2473	51	A ₄₀	0.0199	8.9168	0.6430	3.6778	20
A ₁₅	0.0168	3.1769	0.5448	1.5538	48	A ₄₁	0.0149	3.4881	0.4816	1.6205	47
A ₁₆	0.0257	73.4544	0.8339	25.9353	5	A ₄₂	0.0195	12.1876	0.6333	4.8125	14
A ₁₇	0.0206	9.9250	0.6674	4.0525	19	A ₄₃	0.0179	6.1759	0.5811	2.6591	33
A ₁₈	0.0275	102.3109	0.8900	35.7670	3	A ₄₄	0.0112	3.8486	0.3620	1.6570	46
A ₁₉	0.0163	7.8827	0.5272	3.2162	25	A ₄₅	0.0217	16.5553	0.7038	6.3929	12
A ₂₀	0.0183	4.5680	0.5927	2.0936	39	A ₄₆	0.0179	6.3353	0.5803	2.7150	30
A ₂₁	0.0229	31.1944	0.7410	11.4614	7	A ₄₇	0.0193	8.6230	0.6239	3.5584	22
A ₂₂	0.0211	11.6954	0.6836	4.6859	15	A ₄₈	0.0212	16.1223	0.6859	6.2261	13
A ₂₃	0.0173	4.7555	0.5588	2.1351	37	A ₄₉	0.0188	8.1620	0.6100	3.3846	23
A ₂₄	0.0183	6.2995	0.5922	2.7119	31	A ₅₀	0.0184	7.4947	0.5962	3.1380	26
A ₂₅	0.0217	16.6040	0.7023	6.4083	11	A ₅₁	0.0153	4.6157	0.4968	2.0369	40
A ₂₆	0.0187	6.9754	0.6063	2.9629	27	A ₅₂	0.0098	2.0506	0.3160	0.9770	52